

Dataset Inspection Report

Telco Customer Churn Analysis

Client: ABC Communications Ltd

Consulting Team: AnalystLab Africa Consulting

Dataset: Telco Customer Churn Dataset

Tools: Python, Pandas

1. Dataset Overview

The Telco Customer Churn Dataset contains **7,043 customer records and 21 columns**. It includes information about customer demographics, tenure, subscribed services, contract types, payment methods, charges, and churn status.

The main target variable for this analysis is **churn**, which indicates whether a customer left the company.

2. Dataset Structure

The dataset contains both categorical and numerical variables.

Examples include:

- **Categorical:** gender, Partner, InternetService, Contract, PaymentMethod, Churn
- **Numerical:** SeniorCitizen, tenure, MonthlyCharges, TotalCharges

3. Data Types and Missing Values

During inspection, `TotalCharges` was identified as an object/string rather than a numerical variable. It was converted to numeric using Pandas.

```
df['TotalCharges'] = pd.to_numeric(
    df['TotalCharges'],
    errors='coerce'
)
```

The conversion resulted in **11 missing values** in `TotalCharges`. Because these records represented a very small portion of the dataset, they were removed before analysis.

```
df = df.dropna(subset=['TotalCharges'])
```

4. Duplicate Records

Duplicate records were checked using:

```
df.duplicated().sum()
```

Result: [Insert result]

5. Summary Statistics

Descriptive statistics were generated to examine the numerical variables, including `tenure`, `MonthlyCharges`, and `TotalCharges`.

The analysis showed variation in customer tenure and monthly charges, providing a basis for further exploration of their relationship with customer churn.

6. Data Preparation Summary

The following preparation was completed before analysis:

- Inspected rows and columns.
- Reviewed data types.
- Converted `TotalCharges` to numeric.
- Removed 11 records with missing `TotalCharges`.
- Checked for duplicate records.
- Generated summary statistics.
- Created a numerical `Churn` variable for correlation analysis.

7. Conclusion

The inspection confirmed that the dataset was suitable for customer churn analysis after addressing the `TotalCharges` data-type issue and its associated missing values. The cleaned dataset was subsequently used for exploratory analysis, visualization, business insights, and recommendations.